AdventureWorks Cycles — Sales Analysis

Summary Report | Sales & Product Performance Perspective

1. Business Perspective

This analysis focuses on **Sales Analytics**, covering overall sales performance, product performance, product categories and sales territories. The objective is to identify the main revenue drivers and provide practical findings to support management decision-making.

2. Tools and Methodology

DuckDB / SQL: Used for data loading, table relationships, joins, aggregations and core business calculations. SQL was used to calculate revenue, orders, quantity sold, average order value, annual/monthly trends, product performance and territory performance.

Pandas: Used for data-quality and exploratory work rather than repeating the SQL analysis. Missing values, duplicates, data types and the distribution of order values were examined. An IQR-based approach was also used to identify unusually large orders.

Power BI: Used for the final business dashboard. DAX measures were created for Total Revenue, Total Orders, Total Quantity and Average Order Value. Two pages were developed: an Executive Sales Overview and a Product Performance page.

3. Assumptions and Data Preparation

LineTotal from the sales detail table was treated as the sales value for revenue calculations. Orders were counted using distinct SalesOrderID values because one order can contain multiple detail rows. The supplied Excel files contained a formatting inconsistency around SalesOrderID, so the field was standardized during preparation to allow the sales header and detail tables to relate consistently.

The analysis is descriptive and based on the supplied AdventureWorks dataset. Profitability is not inferred from revenue because product cost and margin were not used as the primary performance measures.

Metric	Result
Total Revenue	$109.85M
Total Orders	31K
Total Quantity	275K
Average Order Value	$3.49K

Findings, Implications and Recommendations

4. Key Findings

1. Strong overall sales: AdventureWorks generated approximately **$109.85M** in revenue across around **31K orders** and **275K units**. Average order value was approximately **$3.49K**.

2. 2013 was the strongest year: Revenue increased from 2011 through 2013, reaching roughly **$44M** in 2013, before falling to roughly **$23M** in 2014. The decline is a key area for management investigation.

3. Bikes dominate revenue: The product analysis shows Bikes contributing overwhelmingly more revenue than Components, Clothing and Accessories. Road Bikes and Mountain Bikes are particularly important subcategories.

4. Geographic performance differs: Southwest is the leading visible territory by revenue, followed by Canada and Northwest. This indicates that sales are not evenly distributed across markets.

5. Volume does not tell the whole story: The product scatter analysis shows differences between quantity sold and revenue generated. Management should therefore consider both unit volume and sales value when evaluating products.

5. Recommendations

1. Investigate the 2014 revenue decline by reviewing product demand, territory performance, order volume and changes in product mix.

2. Protect and grow the core bicycle categories, particularly Road Bikes and Mountain Bikes, through demand monitoring, inventory planning and targeted promotions.

3. Compare leading and weaker territories to identify differences in product mix and demand, then adapt successful approaches to weaker markets where appropriate.

4. Use both revenue and quantity when making product decisions. High-volume products should not automatically be treated as the most valuable products without considering sales value.

6. Conclusion

The analysis combines SQL, Pandas and Power BI to provide a focused view of AdventureWorks' sales and product performance. The main business story is that revenue is strongly driven by bicycle products, territory performance varies considerably, and the significant decline after the 2013 peak deserves further investigation. The dashboard allows management to move from overall sales performance into product-level analysis and identify areas requiring attention.